

AAROHAN FORENSICS

[PRIVACY MODE ACTIVE — PERSONAL DATA REDACTED]

1. Executive Summary

Case ID:	CAS-05700798	Status:	OPEN
Analysis Date:	2026-09-04T08:36:23.216995	Overall Threat Score:	100.0 / 100 (CRITICAL)
Explanation Mode:	LAYMAN	Privacy Mode:	ENABLED

Executive Assessment

This email threat investigation evaluated Case CAS-05700798. The overall risk level is assessed as **CRITICAL (100.0/100)**. This report summarizes key findings in plain language to help non-technical stakeholders understand why the email was flagged, what evidence supports the score, and recommended next steps.

Key Assessment Highlights

Risk Factors Identified	7
Risk Mitigating Evidence	1
Coordinated Campaign Match	YES
Extracted IOC Count	2

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2. Preserved Email Evidence

Header Field	Extracted Value (Evidence Preserved)
Sender (From):	adm...@paypal-security.com
Reply-To:	N/A
Recipient (To):	Undisclosed Recipient
Subject:	Urgent: Action Required on Your Account
Email Date:	Thu, 03 Sep 2026 10:15:00 +0000
Message-ID:	
Case ID:	CAS-05700798

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3. Detailed Risk Assessment

Assessment Component	Points / Weight
ML Content Maliciousness Score	99.9
Technical Header & Alignment Flags	35.0
Final Bounded Score (0-100):	100.0 (CRITICAL)

Risk Increaseers (+ Risk)

Factor	Category	Impact
DMARC authentication failed or unverified	Authentication	+15
SPF authentication failed or unverified	Authentication	+10
DKIM cryptographic signature missing or invalid	Authentication	+10
Malicious attachment payload identified	Payload	+40
Flagged malicious indicator (URL: http://paypal-security-update.com)	Intelligence	+35
Flagged malicious indicator (DOMAIN: paypal-security-update.com)	Intelligence	+35
Correlated structural similarity match with historical campaign cluster	Campaign	+15

Risk Reducers (- Risk / Benign Signals)

Factor	Category	Impact
Consistent sender and response routing	Identity	-5

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4. Protocol Authentication & Alignment

Protocol Check	Status	Technical Evidence & Rationale
SPF Check	FAIL	Sender Policy Framework checks if sending IP address is authorized by domain SPF TXT records.
DKIM Signature	FAIL	DomainKeys Identified Mail verifies cryptographic signature integrity of message body and key headers.
DMARC Policy	FAIL	Domain-based Message Authentication alignment enforcement requiring valid SPF or DKIM alignment.
Reply-To Alignment	ALIGNED	Compares sender domain with Reply-To domain to detect response redirection attack vectors.

Plain Language Explanation for Non-Technical Reviewers

Authentication Summary:

- **SPF:** Confirms if the computer sending this email was approved by the owner of the sender's domain name.
- **DKIM:** A digital stamp confirming the email contents were not altered on the way to your inbox.
- **DMARC:** The domain's rulebook instructing receiving email servers how to handle unverified messages.
- **Reply-To Consistency:** Ensures that if you hit 'Reply', your message goes to the same company shown in the 'From' address.

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5. Deep Forensic Analysis

MIME Boundaries & Parts

- Boundary: =====2646269858457816348==

Sanitized Message Body Excerpt (First 1,000 characters)

Dear Customer, We detected unusual activity on your account. Please click the link below to verify your identity: <http://paypal-security-update.com/login-verify> Additionally, we have attached the latest invoice report for your reference. Regards, Security Team

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6. Network Routing Timeline

Hop #	IP Address	HELO / Server	Location / ISP	Timestamp
1	45.33.32.156	unknown-attacker.net	Fremont, United States, Akamai Technologies, Inc.	2026-09-03T10:14:50+00:00
2	142.250.190.46	gateway.company.com	Chicago, United States, Google LLC	2026-09-03T10:15:00+00:00

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7. Extracted Indicators of Compromise (IOCs)

Type	Indicator Value	Intelligence Status / Verdict
URL	http://paypal-security-update.com	MALICIOUS MATCH
DOMAIN	paypal-security-update.com	MALICIOUS MATCH

Lookalike Domain Analysis

- **Lookalike Flag:** {'type': 'CAMPAIGN_MATCH', 'case_number': 'CAS-59959FC1'}
- **Lookalike Flag:** {'type': 'CAMPAIGN_MATCH', 'case_number': 'CAS-38B1E128'}
- **Lookalike Flag:** {'type': 'CAMPAIGN_MATCH', 'case_number': 'CAS-09D0E442'}
- **Lookalike Flag:** {'type': 'CAMPAIGN_MATCH', 'case_number': 'CAS-1BD9AF8F'}
- **Lookalike Flag:** {'type': 'CAMPAIGN_MATCH', 'case_number': 'CAS-4F37BFD8'}
- **Lookalike Flag:** {'type': 'CAMPAIGN_MATCH', 'case_number': 'CAS-8EBD76D9'}
- **Lookalike Flag:** {'type': 'CAMPAIGN_MATCH', 'case_number': 'CAS-ECD0053D'}
- **Lookalike Flag:** {'type': 'CAMPAIGN_MATCH', 'case_number': 'CAS-0C1B1BAC'}
- **Lookalike Flag:** {'type': 'CAMPAIGN_MATCH', 'case_number': 'CAS-3C3C8FCA'}
- **Lookalike Flag:** {'type': 'CAMPAIGN_MATCH', 'case_number': 'CAS-76251B5F'}
- **Lookalike Flag:** {'type': 'CAMPAIGN_MATCH', 'case_number': 'CAS-E1135FCB'}

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8. Structural Locality Sensitive Hashing (TLSH)

Locality Sensitive Hash (TLSH Signature):

T143D02B0FEA6933B141D38858782E4E5CD185E52015D2E6A86606111E30407DD7D724DB

Campaign Cluster Status:

COORDINATED CAMPAIGN MATCH DETECTED

Mathematically similar email structures were found in historical cases, indicating an active threat actor campaign cluster.

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9. Attack Relationship Graph Topology

Total STIX Graph Objects: 9 Nodes, 0 Relationships

Identified Graph Nodes

Node ID / Label	Type
email-1	observable-email
ip-45.33.32.156	observable-ip
hop-ip-1-45.33.32.156	observable-ipv4-addr
loc-1-United States	observable-location
hop-ip-2-142.250.190.46	observable-ipv4-addr
loc-2-United States	observable-location
ind-0	indicator-malicious-url
ind-1	observable-malicious-domain

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10. Digital Chain of Custody

Custody Attribute	Verifiable Value
Case Identifier:	CAS-05700798
Original .EML SHA-256 Hash:	fa708b67e2dce42cb79be2d86390e47b06b42396bf5b893f0477e6479495212f
TLSH Structural Hash:	T143D02B0FEA6933B141D38858782E4E5CD185E52015D2E6A86606111E30407DD7D724DB
Analysis Timestamp:	2026-09-04T08:36:23.216995
Case Status:	OPEN
Preservation Integrity:	Raw .EML evidence preserved without modification.

Evidentiary Integrity Notice:

This report is generated from raw email evidence preserved by the AAROHAN threat forensics engine. Privacy Mode redactions apply only to presentation representations and do not modify the underlying stored SHA-256 evidence hash or original header data.